

LETTERS TO THE EDITOR

Plaque Versus Protons for Ocular Melanoma

In Regard to Trofimov et al



To the Editor:

We read with great interest the recent publication of Trofimov et al comparing the dose distributions of two commonly used radiation treatment modalities for uveal melanoma: plaque brachytherapy and single beam proton therapy.¹ We congratulate the authors for this considerable research that will enable us to indirectly observe the potential side effects of these two techniques, which have similar oncological outcomes. However, we would like to draw attention to a few issues related to the subject.

First, although plaque brachytherapy and proton radiation therapy are modalities that have been used in the treatment of ocular melanoma for a long time, hypofractionated stereotactic radiation therapy is also used by an increasing number of centers. The inclusion of this technique in this study would help readers compare the dose distributions of all 3 techniques.

A second point we would suggest focusing on is vitreous humor (VH). One of the main functions of the VH is to regulate intraocular oxygen pressure. Despite this critical role, VH is not taken into account in the irradiation process.^{2,3} Deterioration of the VH due to aging, vitrectomy, or other causes decreases its ability to consume oxygen, resulting in a decrease in the oxygen gradient. Current evidence has shown that disruption in the oxygen gradient induces cataract due to oxidation of lens proteins exposed to high oxygen and glaucoma because of damage to trabecular meshwork cells due to the increased oxidative stress.^{2,4-8}

The biophysical properties of VH are critical for the continuation of its functions. In a previous study, our group showed that radiation therapy causes a significant decrease in the expressions of type II, V, and XI collagens in VH. Electron microscopic examinations revealed disrupted collagen fibrillar ultrastructure and dispersed collagen fragments in the VH exposed to radiation.⁹ Also, another study suggested that radiation exposure to the VH affects lens metabolism and consequently increases the risk of cataractogenesis.¹⁰

As expected, the present study ignored the VH in dosimetric calculations. In most of the study scenarios, proton therapy appears superior regarding almost all critical structures. The images in Fig. 1 and graphics of the reference case presented in Fig. 2 show that brachytherapy irradiates almost the entire globe volume, and it is observed that the whole eye, and indirectly the VH, is better preserved by proton therapy.

In conclusion, although the present article explores an important issue and yields significant results, other, newer techniques deserve similar dosimetric comparisons. In addition, dosimetric calculations for the VH, whose importance is almost always ignored in terms of eye health and vision, may reveal additional significant findings and provide a different perspective for researchers and clinicians interested in this issue.

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